Aaryan Kushawaha
Sec. :- “A”

Q. Design a code Converter that receives input in BCD and Generate Output in Gray Code .

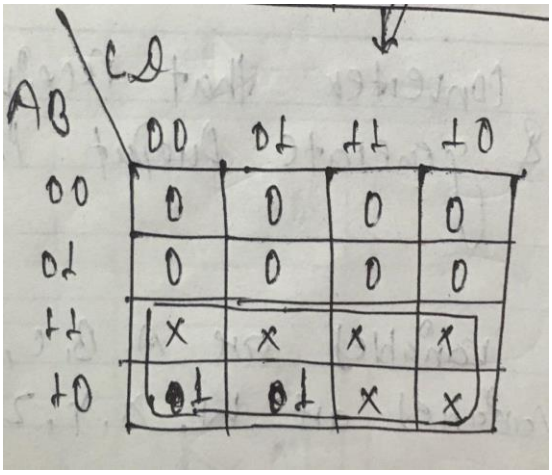
Ans :-

The BCD code is basically 8 4 2 1 code and the conversion of 4 – bit input BCD code (ABCD) into the Gray Code output (WXYZ) as show in Truth Table

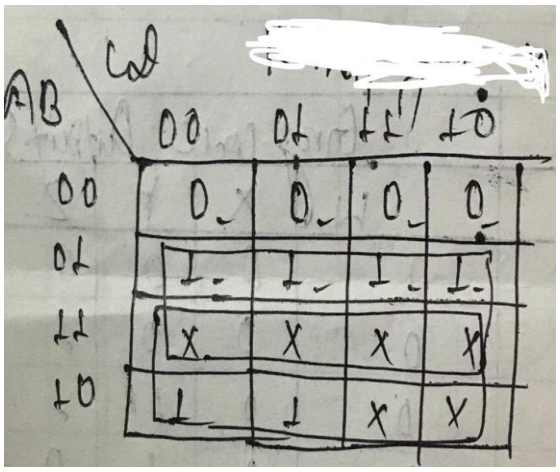
Truth Table :-

Binary Code (Input)				Gray Code (Output)			
A	*B*	*C*	*D*	*W*	*X*	*Y*	*Z*
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1

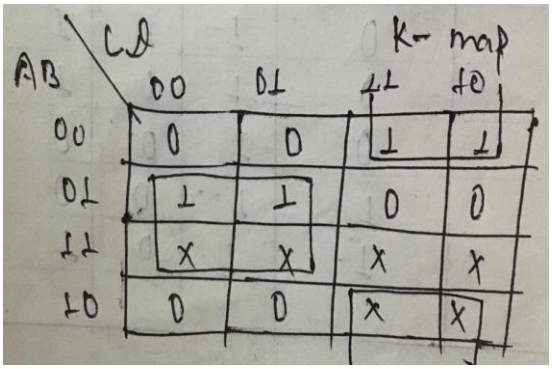
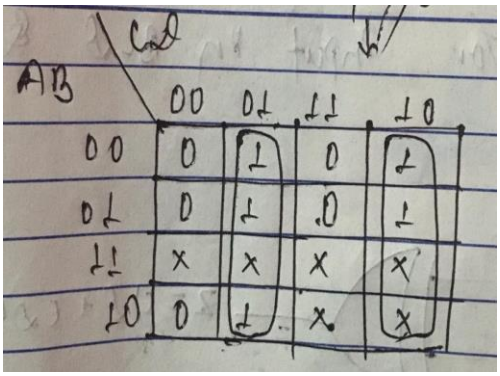
K – Map For W



K-Map for X



K – Map For Y



K – Map For Z